

ANALYSIS REPORT: CUSTOMER SALES & CHURN

ANALYSIS

Project: Advanced Data Manipulation Using Pandas

Program: The Developer Arena Internship — Week 5

Author: Sugam Sagar

1. Executive Summary

This report analyzes two datasets — a **customer churn dataset** (500 customers) and a **sales transaction dataset** (100 orders) — to understand revenue performance, product and customer trends, and the relationship between contract type and customer churn.

Key headline figures from the sales data:

Metric	Value
Total Revenue	\$12,365,048
Total Orders Analyzed	100
Total Unique Customers (Sales)	100
Average Order Value	\$123,650
Top Customer	CUST016 — \$373,932 in sales

2. Datasets

Dataset	Rows	Columns	Description
customer_churn.csv	500	9	Customer tenure, contract type, monthly/total charges, payment method, churn flag
sales_data.csv	100	7	Order-level date, product, quantity, price, customer ID, region, total sales

3. Methodology

- Load & Inspect** — Both CSVs were loaded into Pandas DataFrames and inspected using `.info()` / `.head()` to confirm structure and data types.
- Standardize & Merge** — The `Customer_ID` column in the sales data was renamed to `CustomerID`, and the churn dataset's ID format was transformed to align with the sales dataset before merging with `pd.merge(..., how='left')`.
- Feature Engineering** — The `Date` column was converted to `datetime`, and `Month/Year` fields were derived for time-based aggregation.

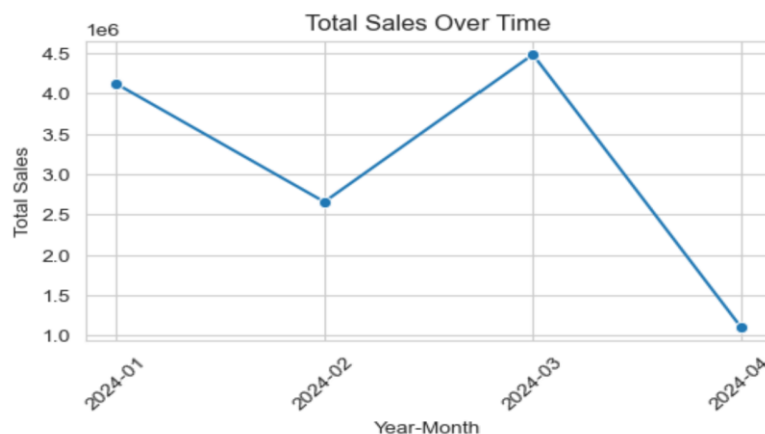
4. **Aggregation** — `groupby()` and `pivot_table()` were used to summarize sales by month, product, customer, and region, and to summarize churn/charges by contract type.
5. **KPI Calculation** — Total revenue, unique customers, average order value, and top customer were computed from the merged dataset.
6. **Visualization** — Seaborn/Matplotlib charts were produced for each aggregation to support visual interpretation.

4. Findings

4.1 Sales Trend Over Time

Year-Month Total Sales

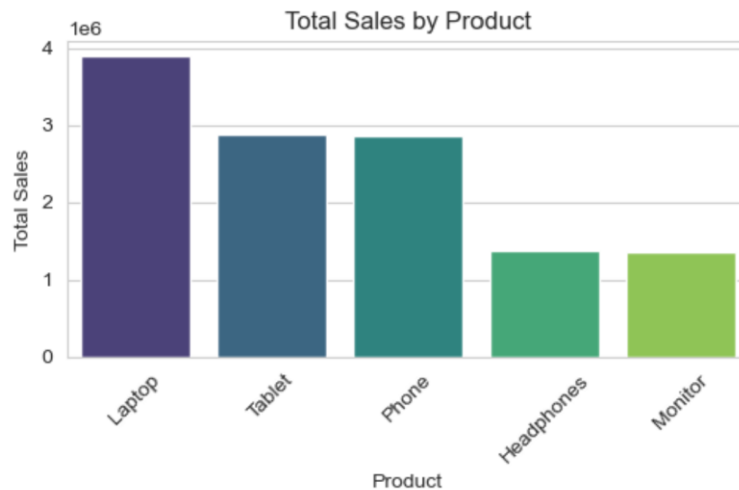
2024-01	\$4,120,524
2024-02	\$2,656,050
2024-03	\$4,485,006
2024-04	\$1,103,468



Sales peaked in **March 2024** and dropped sharply in **April 2024**. The April figure is based on a partial month of data (the dataset ends early in April), so the decline likely reflects incomplete data rather than a genuine demand drop.

4.2 Sales by Product

Product	Total Sales
Laptop	\$3,889,210
Tablet	\$2,884,340
Phone	\$2,859,394
Headphones	\$1,384,033
Monitor	\$1,348,071

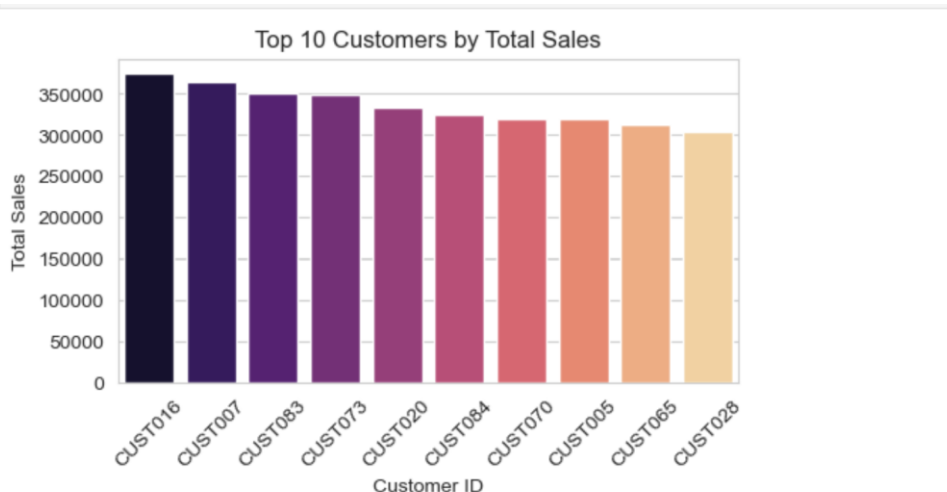


Laptops are the top revenue-generating product, contributing roughly **31%** of total sales, followed closely by tablets and phones. Monitors and headphones are the weakest performers by revenue.

4.3 Top 10 Customers by Sales

Rank Customer ID Total Sales

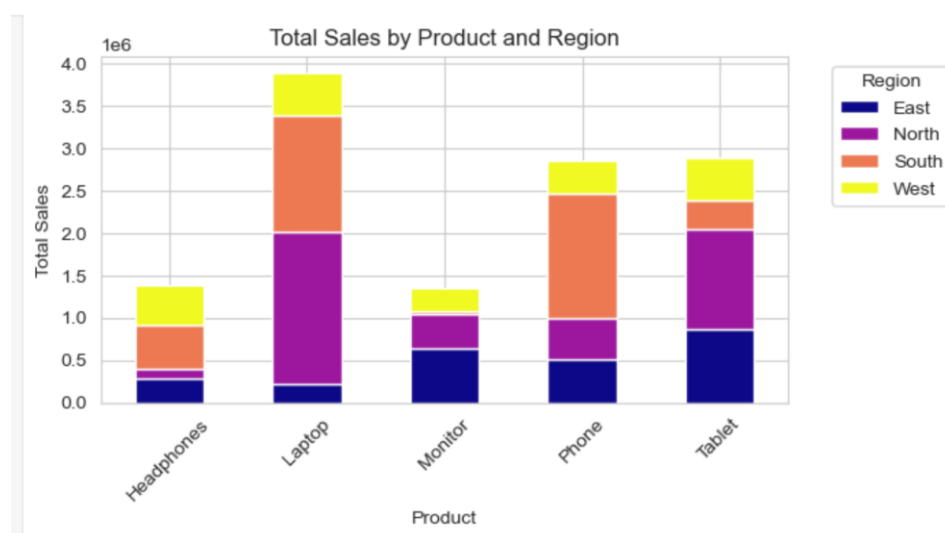
1	CUST016	\$373,932
2	CUST007	\$363,870
3	CUST083	\$350,888
4	CUST073	\$349,510
5	CUST020	\$333,992
6	CUST084	\$324,144
7	CUST070	\$318,762
8	CUST005	\$318,680
9	CUST065	\$312,564
10	CUST028	\$304,465



The top 10 customers collectively account for over **\$3.35M**, roughly **27%** of total revenue — a strong candidate list for loyalty or account-management programs.

4.4 Sales by Product and Region

Product	East	North	South	West
Headphones	288,361	107,091	512,168	476,413
Laptop	221,946	1,798,206	1,373,120	495,938
Monitor	642,870	397,100	39,924	268,177
Phone	506,828	489,284	1,471,428	391,854
Tablet	859,634	1,191,954	341,212	491,540



Laptops dominate in the **North** region, while phones perform best in the **South**. Monitors underperform almost everywhere, particularly in the **South**, suggesting a regional demand or availability gap worth investigating.

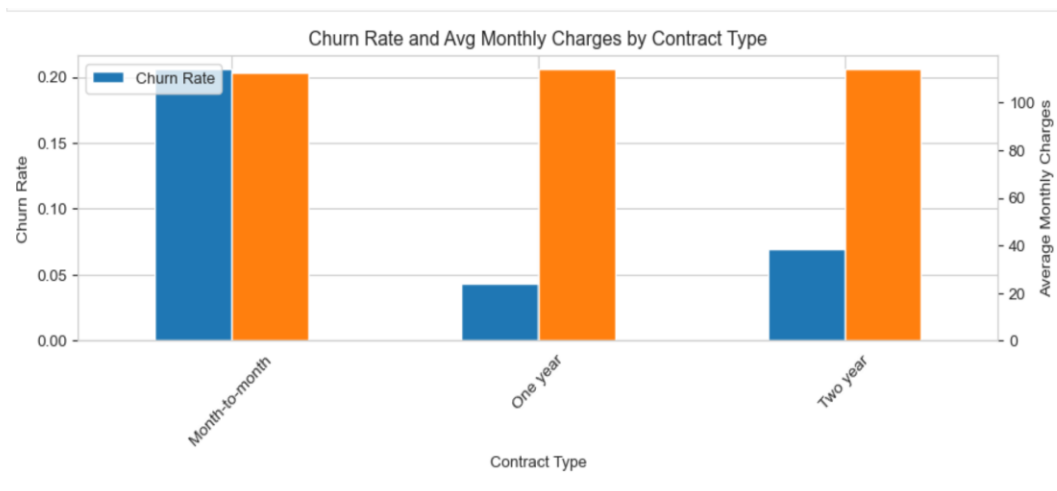
4.5 Churn Rate & Monthly Charges by Contract Type

Contract Type	Churn Rate	Avg. Monthly Charges
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Month-to-month	20.6%	\$112.83
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One year	4.3%	\$114.01
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Two year	6.9%	\$114.11
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Customers on **month-to-month contracts churn nearly 5x more** than those on one-year contracts, despite paying almost the same average monthly charge. This indicates that **contract commitment**, not price, is the dominant driver of retention in this dataset.

5. Data Quality Note

During the merge step, standardizing IDs with `x.replace('C', 'CUST')` converts an ID like `C00001` into `CUST00001` (5 digits), while the sales dataset uses the format `CUST001` (3 digits). Because the digit counts don't match, **the merge produced no matches** — all churn-related columns (Tenure, Contract, Churn, etc.) came through as `NaN` in `merged_df`.

As a result, insights in Sections 4.1–4.4 (derived from `merged_df`) reflect sales data only, while Section 4.5 (derived directly from `customer_churn_df`) is unaffected and reliable.

Recommendation: fix the ID-standardization logic (e.g., zero-pad to a consistent width, such as `'CUST' + x.lstrip('C').zfill(3)`) and re-run the merge to unlock true cross-dataset insights such as "average churn rate of high-value sales customers."

6. Recommendations

1. **Fix the CustomerID standardization logic** so churn and sales data can be reliably joined, enabling deeper insights (e.g., do high-spending customers churn more or less?).
2. **Investigate the April 2024 sales drop** to confirm whether it reflects a partial month of data or an actual decline.
3. **Prioritize retention offers for month-to-month customers**, given their churn rate is significantly higher than longer-term contract holders.

4. **Review monitor product-market fit**, especially in the South region, where sales are disproportionately low.
5. **Build a loyalty program around top 10 customers**, who contribute a disproportionate share of revenue.

7. Conclusion

This analysis demonstrates practical use of Pandas for merging, reshaping, and summarizing multi-source business data. While a data-standardization issue limited the depth of the combined churn/sales analysis, the sales data alone surfaced clear, actionable trends in product performance, regional demand, and top customers — and the churn data independently confirmed that contract length is a strong predictor of customer retention.

Report generated as part of Week 5 of The Developer Arena Internship.